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Lab 02

Inverted Pendulum

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0. Introduction

The purpose of the following report is to tackle one of the classical problems in the discipline of dynamic control of systems: the inverted pendulum on a moving cart. The methodology followed will involve the analytical derivation of the governing equations and corresponding state-space representations, which will help us characterize the system dynamics, and test its response to various inputs on MATLAB® and Simulink®.

We will start by analyzing the system as a normal pendulum hinged to a cart, and we will make comparisons between the nonlinear model and the linearized model.

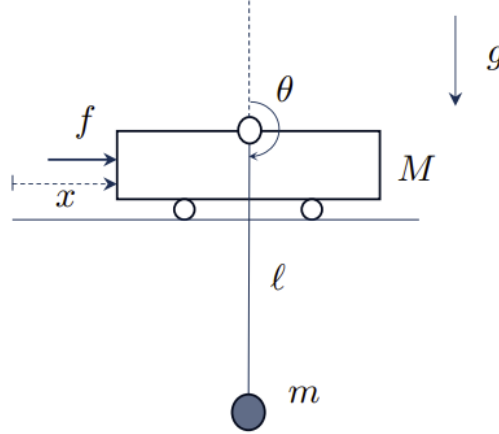


Figure 1: Normal pendulum hinged to the moving cart

Then we will analyze the proper inverted pendulum on a cart, an unstable system, and we will evaluate several control strategies aimed at stabilizing it. Initially, a proportional controller, followed by a proportional-derivative controller, and finally, we will extend our solution to a full state-feedback control.

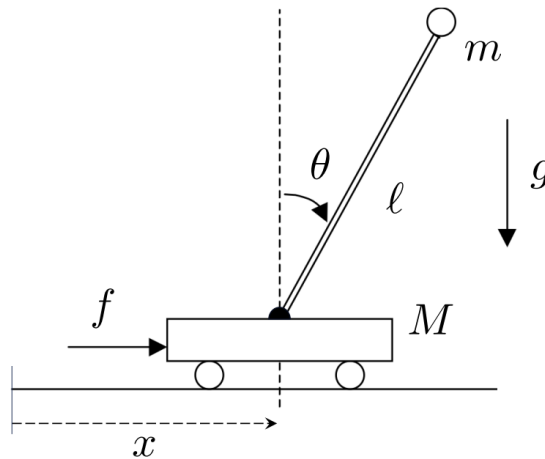


Figure 2: Inverted Pendulum hinged to the moving cart

1. Part 1

1.1. Task 1.1 – Mathematical model

A - Second-order nonlinear EOM of the system

To study the movement of the system we start by writing the Lagrangian equation based on the D.o.F. x and θ :

$$\mathcal{L} = T - V \quad (1)$$

The kinetic energy of the system is expressed as:

$$\begin{aligned} T &= T^c - T^p \\ T^c &= \frac{1}{2} M \dot{x}^2 \end{aligned}$$

Determining the position of the center of mass of the pendulum (considering $y = 0$ in the cart position):

$$\begin{cases} x_m = x + \ell \sin \theta \\ y_m = \ell \cos \theta \end{cases}$$

Finding the velocity of the center of mass:

$$\begin{cases} \dot{x}_m = \frac{dx_m}{dt} = \dot{x} + \ell \dot{\theta} \cos \theta \\ \dot{y}_m = \frac{dy_m}{dt} = -\ell \dot{\theta} \sin \theta \end{cases}$$

$$T^p = \frac{1}{2} m (\dot{x}_m^2 + \dot{y}_m^2) = \frac{1}{2} m (\dot{x}^2 + \ell^2 \dot{\theta}^2 \cos^2 \theta + 2\dot{x}\ell\dot{\theta}\cos\theta + \ell^2 \dot{\theta}^2 \sin^2 \theta) = \frac{1}{2} m (\dot{x}^2 + \ell^2 \dot{\theta}^2 + 2\dot{x}\ell\dot{\theta}\cos\theta)$$

$$T = T^c + T^p = \frac{1}{2} (M + m) \dot{x}^2 + \frac{1}{2} m \ell^2 \dot{\theta}^2 + m \ell \dot{x} \dot{\theta} \cos \theta$$

The potential energy of the system is expressed as:

$$V = V^c + V^p$$

$$V^c = 0$$

$$V^p = mgh_m = mgl\cos\theta$$

$$V = V^c + V^p = mgl\cos\theta$$

Thus, the Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2} (M + m) \dot{x}^2 + \frac{1}{2} m \ell^2 \dot{\theta}^2 + m \ell \dot{x} \dot{\theta} \cos \theta - mgl\cos\theta \quad (2)$$

And the Lagrange's equation of the system can be written as follows:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) - \frac{\partial \mathcal{L}}{\partial q_k} = Q_k \quad (3)$$

Given the D.o.F x and θ :

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} &= Q_\theta \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} &= Q_x \end{aligned}$$

We first differentiate with respect to x :

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \dot{x}} &= (M + m)\dot{x} + m\ell\dot{\theta}\cos\theta \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) &= (M + m)\ddot{x} + m\ell\ddot{\theta}\cos\theta - m\ell\dot{\theta}^2\sin\theta \\ \frac{\partial \mathcal{L}}{\partial x} &= 0\end{aligned}$$

The generalized force Q_x associated with the coordinate x is derived using the principle of virtual work:

$$\delta W = \tau_{\text{friction}} \delta x + f \delta x = (-c\dot{x} + f) \delta x = Q_x \delta x \quad (4)$$

$$\Rightarrow Q_x = f - c\dot{x}$$

where c is the viscous damping coefficient.

The equation becomes:

$$(M + m)\ddot{x} + m\ell\ddot{\theta}\cos\theta - m\ell\dot{\theta}^2\sin\theta = f - c\dot{x}$$

Then we differentiate with respect to θ :

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} &= m\ell^2\dot{\theta} + m\ell\dot{x}\cos\theta \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) &= m\ell^2\ddot{\theta} + m\ell\ddot{x}\cos\theta - m\ell\dot{x}\dot{\theta}\sin\theta \\ \frac{\partial \mathcal{L}}{\partial \theta} &= -m\ell\dot{x}\dot{\theta}\sin\theta + mg\ell\sin\theta\end{aligned}$$

The generalized force Q_θ associated with the coordinate θ is derived using the principle of virtual work:

$$\delta W = \tau_{\text{friction}} \delta\theta = (-b\dot{\theta}) \delta\theta = Q_\theta \delta\theta \quad (5)$$

$$\Rightarrow Q_\theta = -b\dot{\theta}$$

where b is the viscous damping coefficient.

The equation becomes:

$$m\ell^2\ddot{\theta} + m\ell\ddot{x}\cos\theta - mg\ell\sin\theta = -b\dot{\theta}$$

The resulting equations are:

$$\begin{aligned}(M + m)\ddot{x} + m\ell\ddot{\theta}\cos\theta - m\ell\dot{\theta}^2\sin\theta &= f - c\dot{x} \\ m\ell^2\ddot{\theta} + m\ell\ddot{x}\cos\theta - mg\ell\sin\theta &= -b\dot{\theta}\end{aligned}$$

(6)

B - Nonlinear state-space formulation

- State vector:

$$\mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{Bmatrix}$$

- Selecting the force as input:

$$u = f$$

- Selecting as output x_1 which represents the cart horizontal displacement and x_3 which corresponds to the pendulum angular position with respect to the vertical axis:

$$\mathbf{y} = \begin{bmatrix} x \\ \theta \end{bmatrix} = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix}$$

Our next objective is to explicit $\ddot{x}$ and $\ddot{\theta}$ from the EOM:

$$\begin{bmatrix} M + m & m\ell\cos x_3 \\ m\ell\cos x_3 & m\ell^2 \end{bmatrix} \begin{bmatrix} \dot{x}_2 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} u - cx_2 + m\ell x_4^2 \sin x_3 \\ -bx_4 + mgl \sin x_3 \end{bmatrix} \quad (7)$$

We can solve the system by inverting the mass matrix.

Let:

$$D = (M + m)m\ell^2 - (m\ell\cos x_3)^2$$

And after the multiplication of the inverse of the mass matrix with the vector on the right-hand side, the accelerations are given by:

$$\begin{cases} \ddot{x} = \dot{x}_2 = \frac{1}{D} [m\ell^2(u - cx_2 + m\ell x_4^2 \sin x_3) - m\ell \cos x_3 (-bx_4 + mgl \sin x_3)] \\ \ddot{\theta} = \dot{x}_4 = \frac{1}{D} [-m\ell \cos x_3 (u - cx_2 + m\ell x_4^2 \sin x_3) + (M + m)(-bx_4 + mgl \sin x_3)] \end{cases} \quad (8)$$

The nonlinear state-space representation of the inverted pendulum system can now be fully expressed in first-order form.

Compact version:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} x_2 \\ \ddot{x}(\mathbf{x}, u) \\ x_4 \\ \ddot{\theta}(\mathbf{x}, u) \end{bmatrix} \quad (9)$$

1.2. Task 1.2 – Nonlinear response to a force applied on the cart (MATLAB[®] code)

In this task we want to simulate and analyze the non-linear open-loop response of an inverted pendulum system subject to an external force applied on the cart. Our analysis was focused on investigating how different impulse force amplitudes affect the dynamic behavior of the pendulum angle and the cart position.

As shown in **Task 1.1**, the nonlinear equations of motion are:

$$\begin{cases} (M + m)\ddot{x} + m\ell\ddot{\theta}\cos\theta - m\ell\dot{\theta}^2\sin\theta = f - c\dot{x} \\ m\ell^2\ddot{\theta} + m\ell\ddot{x}\cos\theta - mgl\sin\theta = -b\dot{\theta} \end{cases}$$

These were rearranged into matrix form (7) and solved numerically in MATLAB[®] using the `ode45` solver, chosen due to its robustness and efficiency when dealing with non-stiff ordinary differential equations.

We observed that, for long simulation horizons ($t_f > 100$), using `ode45` with a time span definition $[0 \ t_f]$ can lead to numerical instabilities or solution blow-up. By instead specifying a denser time vector such as $[0:0.001:t_f]$, the solver is indirectly constrained to evaluate the solution more frequently, resulting in improved numerical stability and accuracy.

Although a time vector such as $[0:0.01:20]$ would already provide a sufficiently good resolution, we opted for a denser vector, $[0:0.001:20]$, as it does not significantly affect computation time while clearly offering a more accurate solution.

The system was implemented in a custom function `ipend_ode.m`, which accepts a time-varying force function as input. To evaluate the response under different excitation levels, a 1-second rectangular impulse force was applied (from $t = 1$ to $t = 2$ seconds), with amplitudes of 1 N, 5 N and 25 N, implemented by a parametric force function $f_{pulse}(t, tspan, amp)$.

The initial condition corresponds to the equilibrium position: $x = 0$, $\theta = \pi$, and zero initial velocities.

For each simulation, the system states were integrated over a 20-second interval.

For each input force amplitude, the cart position and pendulum angle were plotted over time.

The results are shown below:

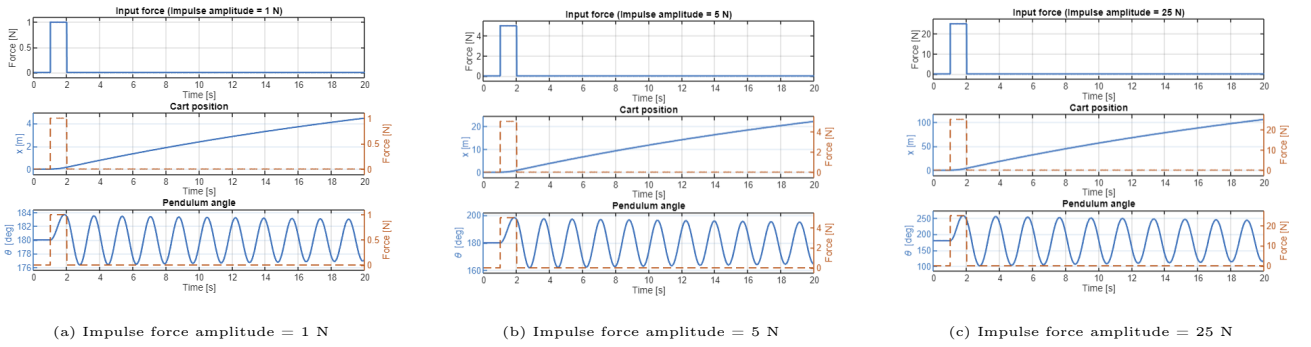


Figure 3: Nonlinear open-loop response of the inverted pendulum system for different impulse force amplitudes. Each plot shows the pulse force, the cart position and pendulum angle over time for a 1-second impulse applied from $t = 1$ s to $t = 2$ s.

The pendulum starts from the stable equilibrium ($\theta = 180^\circ$), with angles measured clockwise from the vertical.

For an impulse of 1 N the cart shows limited motion, and the pendulum manifests small and regular oscillations around equilibrium angle.

With 5 N the cart displacement and pendulum amplitude increase, revealing clear nonlinear behavior.

At 25 N the response becomes strongly nonlinear: the cart travels farther and the pendulum exhibits large oscillations between approximately 110° and 250° , without completing a full rotation.

1.3. Task 1.3 – Nonlinear response to a force applied on the cart (Simulink[®] models)

A - Model based on integral block approach

In order to implement the Simulink[®] model using integral blocks, it was necessary to employ the non-linear state-space formulation of the equations of motion (EOM). Specifically, we selected the equations in which the cart acceleration $\ddot{x}$ and the angular acceleration of the pendulum $\ddot{\theta}$ are explicitly expressed:

$$\begin{cases} \ddot{x} = \dot{x}_2 = \frac{1}{D} [m\ell^2(u - c\dot{x}_2 + m\ell\dot{x}_4^2\sin x_3) - m\ell\cos x_3(-b\dot{x}_4 + mgl\sin x_3)] \\ \ddot{\theta} = \dot{x}_4 = \frac{1}{D} [-m\ell\cos x_3(u - c\dot{x}_2 + m\ell\dot{x}_4^2\sin x_3) + (M + m)(-b\dot{x}_4 + mgl\sin x_3)] \end{cases}$$

$$D = (M + m)m\ell^2 - (m\ell\cos x_3)^2$$

These accelerations are essential for applying the integral block method, which is based on the successive integration of the system second-order differential equations to compute the system state variables over time.

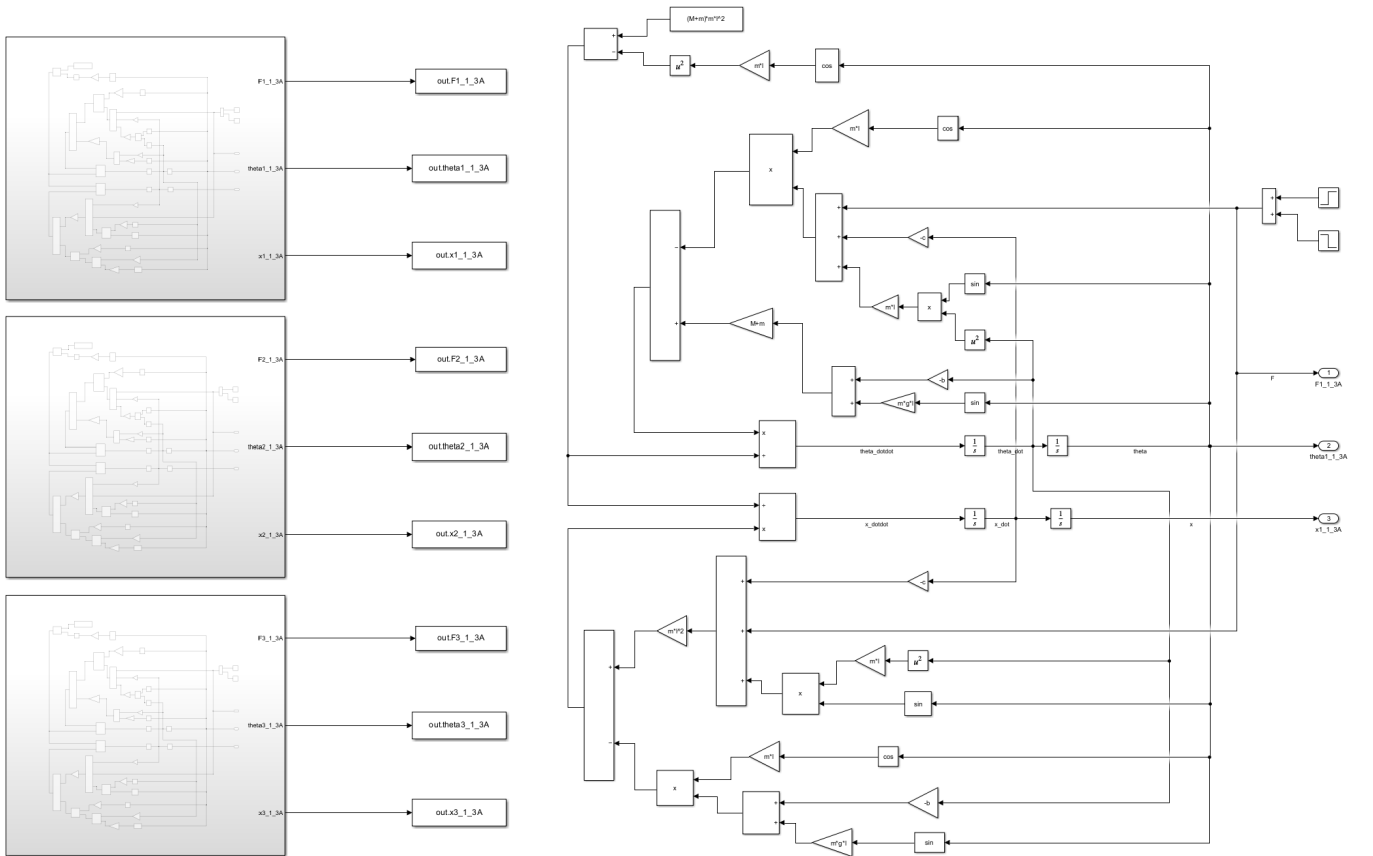
To model the nonlinear response to a force applied to the cart, we chose the simulation parameters and blocks.

Regarding the simulation settings, the Start time was defined as 0.0 s and the Stop time as 20.0 s. A variable-step solver (ode45) was chosen for numerical integration and the Max step size was reduced from the default value of 0.04 to 0.01, forcing the solver to take smaller steps and thus achieve a more accurate and precise solution.

As for the blocks used in the model, the following were selected:

- **Step** blocks (used to apply a 1-second pulse force F — from 1 to 2 s — at varying amplitudes on the cart)
- **Integral** blocks (configured with the initial conditions $x_0 = [0; 0; \pi; 0]$, used to compute the system states over time by integrating the derivatives)
- **Sum**, **Product**, and **Divide** blocks
- **Gain** blocks (used to scale signals according to model parameters or constants)
- **To Workspace** blocks (used to export the input force F , the x , and the pendulum angle θ to MATLAB[®])

To obtain the three outputs (F , x , and θ) for each of the three cases analyzed (amplitudes of 1 N, 5 N, and 25 N), we created three subsystems. Each subsystem utilized the blocks previously mentioned and contains within it blocks such as Outputport to provide output signals, then sent to the To Workspace blocks. Below is the Simulink[®] model and one of the three subsystems:



Having completed the Simulink model based on an integral block approach, three plots were generated, each containing the pulse force plot (1 N, 5 N, and 25 N), the cart position plot, and the pendulum angle plot over a simulation time of 20.0 s. The results confirm that the nonlinear response is consistent and aligns with the previous graphs (see Fig. 3).

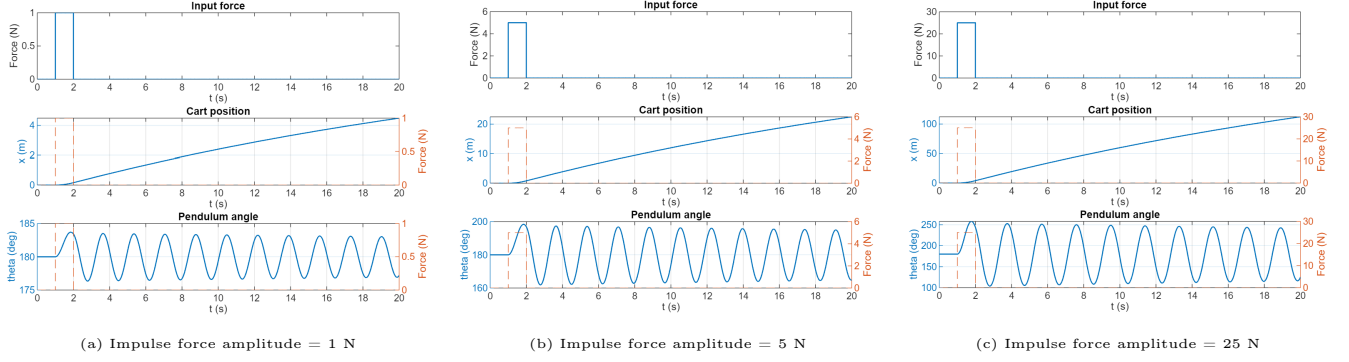


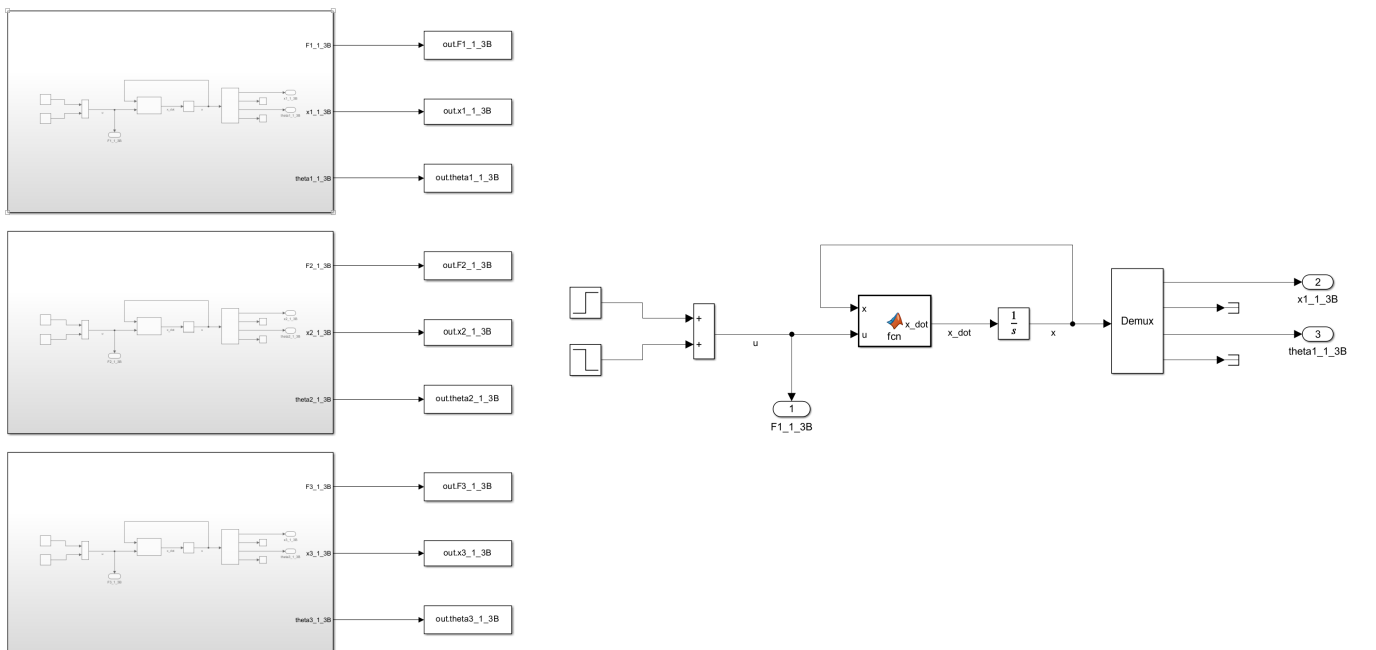
Figure 4: Nonlinear open-loop response derived from a Simulink[®] model based on integral block approach.

B - Model based on M-function

After implementing the previous model, the model based on M-function proved to be more compact and understandable, since it required the use of fewer blocks. The model reproduces the step force that goes directly into the M-function, the M-function creates the state-space formulation and the first derivative of the system, which is subsequently integrated and used as an input for the function itself (the parameters are assigned within the function). We selected the same simulation parameters as in the previous model, using `ode45` as the solver with a Max step of 0.01. The blocks chosen to make the model are as follows:

- **Step** blocks (used to apply a 1-second pulse force F — from 1 to 2 s — with varying amplitudes on the cart)
- **Integral** blocks (configured with the initial conditions $x_0 = [0; 0; \pi; 0]$, used to compute the system states over time by integrating their derivatives)
- **Sum** blocks (used to recreate the step force)
- **Demux** blocks (used to separate the outputs in order to send x and θ to the workspace)
- **Terminator** blocks (used to terminate signals that are not connected to other blocks — in this case, $\dot{x}$ and $\dot{\theta}$)
- **To Workspace** blocks (used to export the outputs to MATLAB[®])

Also in this model, three subsystems were used to obtain the three outputs (F , x , and θ) for the three different force amplitudes of 1 N, 5 N, and 25 N. Shown below is the Simulink[®] model and one of the three subsystems:



After finalizing the Simulink[®] model based on M-function, 3 plots were created with each having the input force F , cart position x and pendulum angle θ . The results were validated by the previous graphical outcomes and were consistent with them (see Fig. 3).

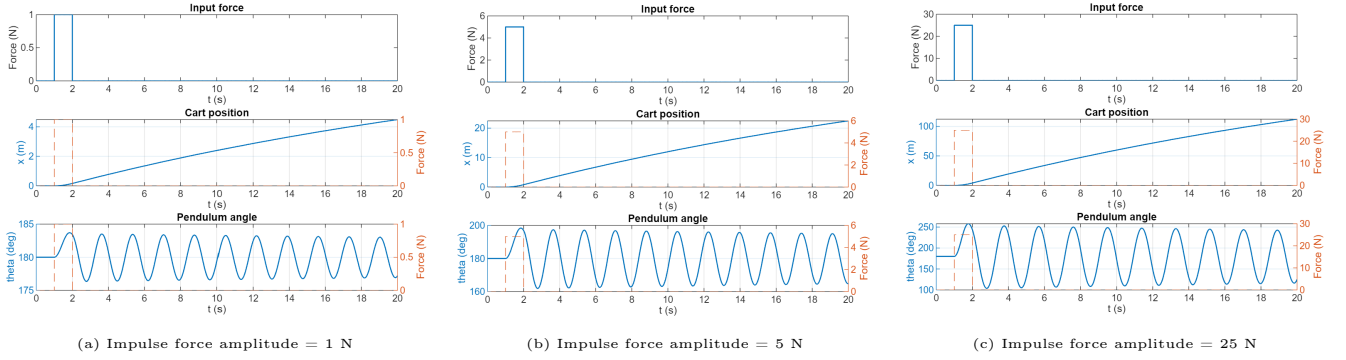


Figure 5: Nonlinear open-loop response derived from a Simulink[®] model based on M-function.

1.4. Task 1.4 – Analysis of the linear(ized) system

A - Derivation of the state-space matrices

The second-order system can be expressed in state-space form as:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u), \quad \mathbf{y} = \mathbf{g}(\mathbf{x})$$

The state vector derived can be written as:

$$\dot{\mathbf{x}} = \begin{bmatrix} f_1(\mathbf{x}, u) \\ f_2(\mathbf{x}, u) \\ f_3(\mathbf{x}, u) \\ f_4(\mathbf{x}, u) \end{bmatrix} = \begin{bmatrix} x_2 \\ \ddot{x}(\mathbf{x}, u) \\ x_4 \\ \ddot{\theta}(\mathbf{x}, u) \end{bmatrix}$$

The accelerations were previously derived (1.3).

We also recall the state, input, and output vectors defined as:

$$\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad u = f, \quad \mathbf{y} = \begin{bmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} x \\ \theta \end{bmatrix} = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix}$$

The aim is to derive the mathematical expression and numerical values of the state-space matrices (A, B, C, D) of the linearized model about the equilibrium point:

$$\mathbf{x}_{eq} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}_{eq} = \begin{bmatrix} 0 \\ 0 \\ \pi \\ 0 \end{bmatrix}$$

The resulting linearized system will take the form:

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}u \end{cases} \quad (10)$$

The state matrix A , input matrix B , output matrix C , and feedthrough matrix D are derived analytically by computing them as follows:

$$A = \left[\begin{array}{cccc} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{array} \right] \bigg|_{x_{eq}}$$

$$B = \left[\begin{array}{c} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \\ \frac{\partial f_3}{\partial u} \\ \frac{\partial f_4}{\partial u} \end{array} \right] \bigg|_{x_{eq}}$$

$$C = \left[\begin{array}{cccc} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \frac{\partial g_1}{\partial x_3} & \frac{\partial g_1}{\partial x_4} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \frac{\partial g_2}{\partial x_3} & \frac{\partial g_2}{\partial x_4} \end{array} \right] \bigg|_{x_{eq}}$$

$$D = \left[\begin{array}{c} \frac{\partial g_1}{\partial u} \\ \frac{\partial g_2}{\partial u} \end{array} \right] \bigg|_{x_{eq}}$$

After executing the derivations and the evaluation at the equilibrium, we get the definitive state-space matrices:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{c}{M} & -\frac{mg}{M} & -\frac{b}{\ell M} \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{c}{M\ell} & -\frac{(M+m)g}{M\ell} & -\frac{(M+m)b}{m\ell^2 M} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ \frac{1}{M\ell} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using the system parameters,

$$M = 3.0 \text{ kg}, \quad m = 0.1 \text{ kg}, \quad \ell = 0.75 \text{ m}, \quad g = 9.8 \text{ m/s}^2, \quad c = 0.1 \text{ N/m/s}, \quad b = 0.001 \text{ Nm/rad/s}$$

we can compute the numerical values of the matrices:

$$A \approx \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -0.0333 & -0.3267 & -0.0004 \\ 0 & 0 & 0 & 1 \\ 0 & -0.0444 & -13.5022 & -0.0184 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0.3333 \\ 0 \\ 0.4444 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

B - Derivation of the transfer functions

The transfer function of a linear time-invariant system in state-space form is given by:

$$G(s) = C(sI - A)^{-1}B + D \quad (11)$$

Since $D = 0$, the expression simplifies to:

$$G(s) = C(sI - A)^{-1}B$$

In our case, the system has one input $u = f$ and two outputs: the x and the pendulum angle θ . We will be therefore computing two transfer functions:

$$\begin{cases} G_1(s) = \frac{X(s)}{U(s)} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} (sI - A)^{-1}B & \text{(cart position)} \\ G_2(s) = \frac{\Theta(s)}{U(s)} = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} (sI - A)^{-1}B & \text{(pendulum angle)} \end{cases} \quad (12)$$

We can compute this otherwise difficult calculations on MATLAB[®] and find the expressions of the two transfer functions:

$$G_1(s) = \frac{m\ell^2 s^2 + bs + gml}{s(M\ell^2 ms^3 + c\ell^2 ms^2 + bms^2 + Mbs^2 + g\ell m^2 s + Mglms + bcs + cg\ell m)}$$

$$G_2(s) = \frac{\ell ms}{M\ell^2 ms^3 + c\ell^2 ms^2 + bms^2 + Mbs^2 + g\ell m^2 s + Mglms + bcs + cg\ell m}$$

Both should share the same denominator, which corresponds to the characteristic polynomial of the state matrix A , but a visible factor of s appears in the denominator of $G_1(s)$ but not in $G_2(s)$. This is due to a common factor s present in both the numerator and denominator of $G_2(s)$, which cancels out algebraically. In contrast, the numerator of $G_1(s)$ does not contain s , so the factor at the denominator remains explicit in the final expression.

Now we can substitute the numerical values into the TFs:

$$G_1(s) = \frac{0.05625s^2 + 0.001s + 0.735}{s(0.16875s^3 + 0.008725s^2 + 2.2786s + 0.0735)}$$

$$G_2(s) = \frac{0.075s}{0.16875s^3 + 0.008725s^2 + 2.2786s + 0.0735}$$

C - Pole-zero maps

Now we can plot the pole-zero map for both TFs, which will be done with the MATLAB[®] plot function `pzmap`. The results are shown in Fig.6:

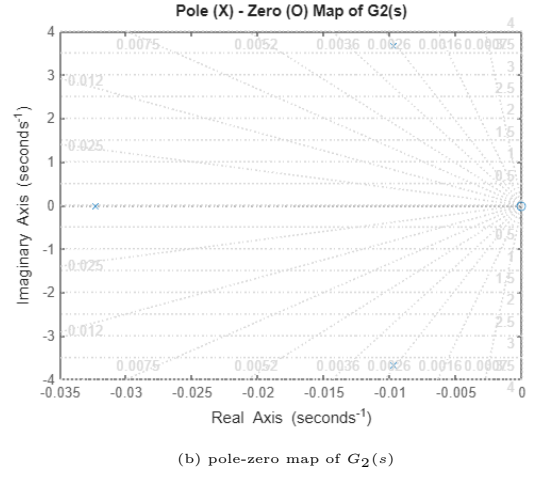
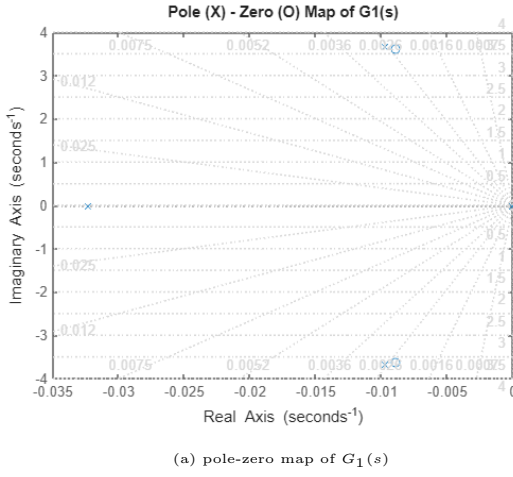


Figure 6: Pole-zero maps for the two TFs.

The poles and zeros of the transfer functions $G_1(s)$ and $G_2(s)$ are essential in determining the system's stability and dynamic behavior. Poles with negative real parts indicate stability, while zeros influence the frequency response. The pole-zero maps reveal that both systems share almost the same poles, with the exception of one additional pole in 0 for $G_1(s)$. In the other hand, $G_2(s)$ has its only zero in 0, while $G_1(s)$ has two not-null.

D - Asymptotic and real Bode diagrams

Having previously derived the expressions of the two transfer functions along with their respective poles and zeros, we can now proceed to plot both the asymptotic and the actual Bode diagrams for each system:

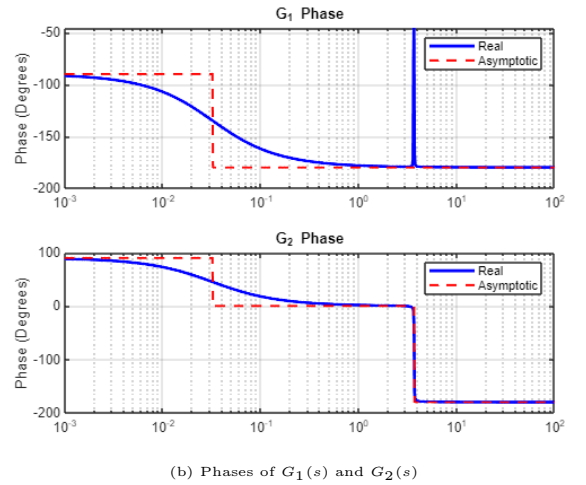
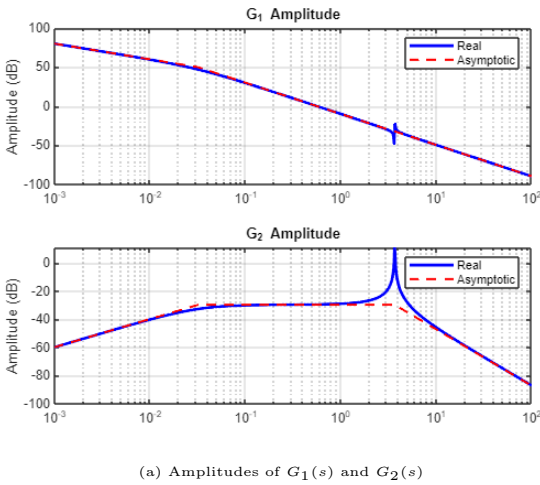


Figure 7: Bode diagrams of the two TFs

1.5. Task 1.5 – Comparison of the nonlinear and linear response (MATLAB[®] code)

In this task, we compare the dynamic responses of the inverted pendulum system under impulsive excitations of varying magnitudes, using both the nonlinear model implemented in Task 1.2 and its linearized counterpart. The goal is to evaluate the validity of the linear approximation and identify when nonlinear effects become significant.

To compute the linear system response to an impulsive force, we used MATLAB[®] `lsim` function.

This function is well-suited for simulating the time response of linear time-invariant (LTI) systems to arbitrary inputs. It avoids the need for manual time-domain integration and efficiently applies the input to the transfer function models $G_1(s)$ and $G_2(s)$ previously obtained in the frequency domain.

Since the linear model describes deviations around this equilibrium, its angular output represents small variations $\Delta\theta$ with respect to $\theta = 180^\circ$. Consequently, to compare it directly with the nonlinear solution—where the absolute angle is integrated over time—the linear result must be offset by 180° in the MATLAB[®] code: $\theta_{\text{linear}}(t) = 180^\circ + \Delta\theta(t)$

This offset ensures that both models are expressed in the same reference frame and can be meaningfully compared.

Results and Discussion

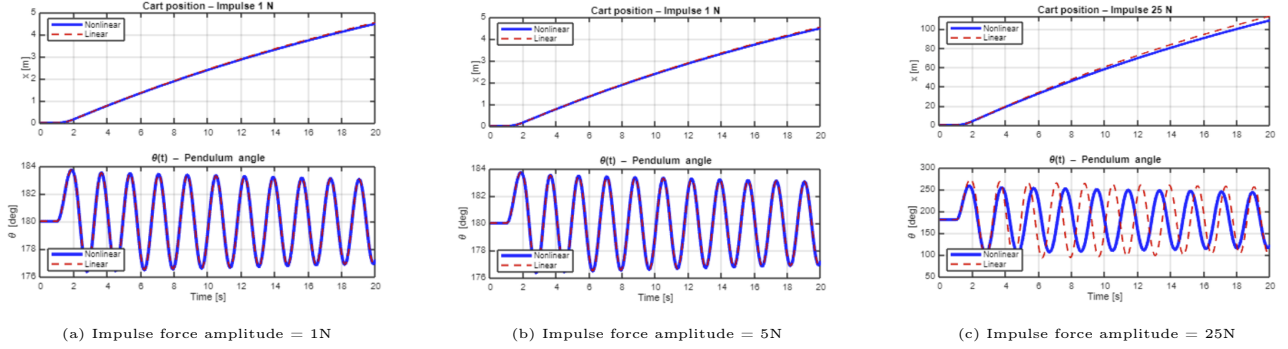


Figure 8: Comparison of Linear and Nonlinear response of the pendulum system for different impulse force amplitudes. Each plot shows the (top) and pendulum angle (bottom) over time for a 1-second impulse applied from $t = 1$ s to $t = 2$ s.

• Cart Position (Top Plot):

The linear and nonlinear cart trajectories show excellent agreement across the entire simulation interval, indicating that the translational dynamics of the cart remain essentially linear even under high-force inputs. The influence of pendulum motion on the cart trajectory does not significantly alter its linearity in this setup.

• Pendulum Angle (Bottom Plot):

Differences between the models become evident in the pendulum angular response. The linear model predicts oscillations of higher amplitude and frequency compared to the nonlinear model. This discrepancy reflects the inability of the linear approximation to capture amplitude-dependent dynamics inherent in the full nonlinear equations. In particular, for large excursions from equilibrium, the angular behavior of the pendulum no longer follows the linearized assumptions, leading to noticeable deviations between the two models.

For lower input amplitudes (1 N and 5 N), these differences are minimal.

The responses are nearly indistinguishable for 1 N, while for 5 N, a slight phase shift appears after approximately 14 seconds.

These results confirm that the linear model offers a valid approximation for small deviations but becomes less accurate for larger angles induced by higher input forces.

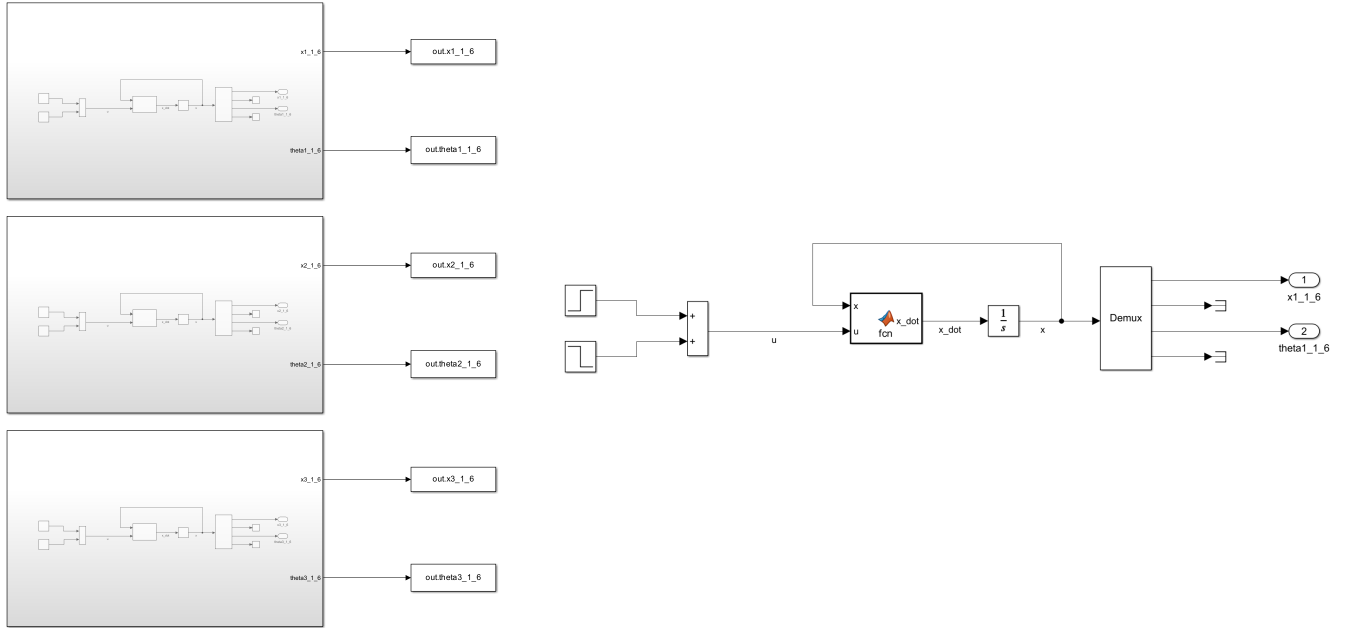
1.6. Task 1.6 – Comparison of the nonlinear and linear response (Simulink[®] models)

To obtain the linear open-loop response in terms of cart position and pendulum angle over the time interval from 0 to 20 seconds, we developed a Simulink[®] model based on an M-function, previously used in **Task 1.3**.

The system employs a function that takes as input the control signal u , derived from step force inputs, and computes the first derivative of the linearized state-space system using the equation $\dot{x} = A(x - x_0) + Bu$. This derivative is then integrated, and the resulting state vector x is fed back into the function. All required parameters and matrices are defined within the M-function. For the simulation settings, the start time was set to 0.0 s and the stop time to 20.0 s. The `ode45` solver was selected for numerical integration, with a Max step size of 0.01. Regarding the block selection, the following components were used:

- **Step** blocks (used to apply a 1-second pulse force F — from 1 to 2 s — with varying amplitudes on the cart)
- **Integral** blocks (configured with the initial conditions $x_0 = [0; 0; \pi; 0]$, used to compute the system states over time by integrating their derivatives)
- **Sum** blocks (used to recreate the step force)
- **Demux** blocks (used to separate the outputs in order to send x and θ to the workspace)
- **Terminator** blocks (used to terminate signals that are not connected to other blocks — in this case, $\dot{x}$ and $\dot{\theta}$)
- **To Workspace** blocks (used to export the outputs to MATLAB[®])

Three subsystems were created with two outputs for each to export the position of the cart x and angle of the pendulum θ to be linked to the blocks to workspace for later use in MATLAB[®]. Below is the Simulink[®] model and one of the three subsystems:



After completing the model, the linear and nonlinear responses obtained from the previous Simulink[®] implementations were plotted to analyze their differences:

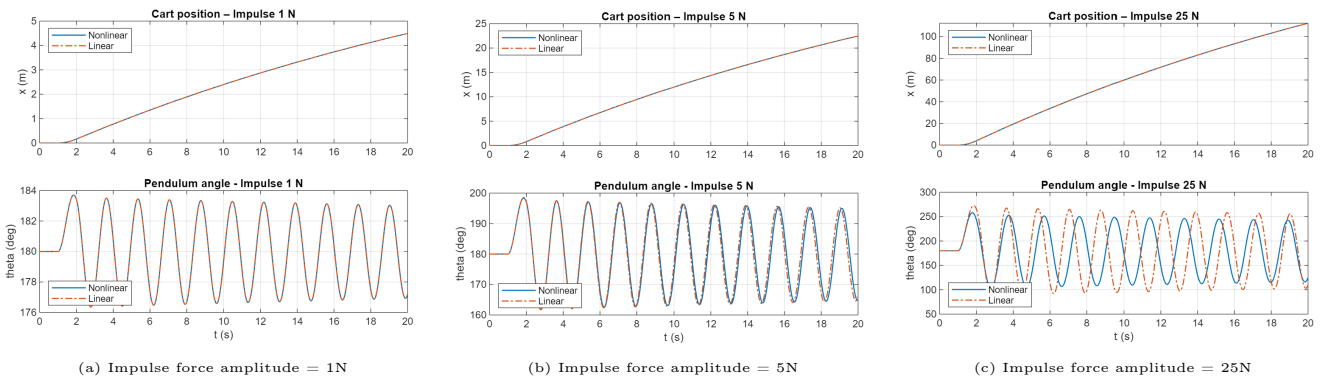


Figure 9: Comparison of Linear and Nonlinear response of the pendulum system for different impulse force amplitudes based on Simulink[®] models. Each plot shows the (top) and pendulum angle (bottom) over time for a 1-second impulse applied from $t = 1$ s to $t = 2$ s.

The simulation results confirm — consistently with those obtained in **Task 1.5** — that the linear model provides a reliable approximation for small deviations from equilibrium, corresponding to low-magnitude forces applied to the cart. However, as the input step force increases, the effects of the system inherent nonlinearity become evident, particularly in the pendulum angular response, which exhibits significant deviations from the linear prediction.

2. Part 2

2.1. Task 2.1 – Mathematical model

We are now requested to study the same system with some differences:

- The pendulum angle θ is considered positive in the counterclockwise direction.
- The friction coefficients b and c are assumed to be zero.
- The input force f is now expressed as the sum of a control force u and a disturbance force d .
- The new equilibrium point considered is $\theta = 0$, therefore the pendulum is pointed upwards, meaning that the equilibrium should now be unstable.

Let's derive the EOM of the system as before considering these differences.

A - Second-order nonlinear EOM of the system

We begin by writing the Lagrangian equation based on the D.o.F. x and θ :

$$\mathcal{L} = T - V \quad (13)$$

The kinetic energy of the system is expressed as:

$$\begin{aligned} T &= T^c + T^p \\ T^c &= \frac{1}{2} M \dot{x}^2 \end{aligned}$$

The position of the center of mass of the pendulum is now (considering $y = 0$ in the cart position):

$$\begin{cases} x_p = x - \ell \sin \theta \\ y_p = \ell \cos \theta \end{cases}$$

The velocity of the center of mass becomes:

$$\begin{cases} \dot{x}_m = \frac{dx_m}{dt} = \dot{x} - \ell \dot{\theta} \cos \theta \\ \dot{y}_m = \frac{dy_m}{dt} = -\ell \dot{\theta} \sin \theta \end{cases}$$

$$T^p = \frac{1}{2} m (\dot{x}_m^2 + \dot{y}_m^2) = \frac{1}{2} m (\dot{x}^2 + \ell^2 \dot{\theta}^2 \cos^2 \theta - 2\dot{x}\ell\dot{\theta}\cos\theta + \ell^2 \dot{\theta}^2 \sin^2 \theta) = \frac{1}{2} m (\dot{x}^2 + \ell^2 \dot{\theta}^2 + 2\dot{x}\ell\dot{\theta}\cos\theta)$$

$$T = T^c + T^p = \frac{1}{2} (M + m) \dot{x}^2 + \frac{1}{2} m \ell^2 \dot{\theta}^2 - m \ell \dot{x} \dot{\theta} \cos \theta$$

The potential energy of the system is still expressed as:

$$V = V^c + V^p$$

$$V^c = 0$$

$$V^p = mgh = mgl \cos \theta$$

$$V = V^c + V^p = mgl \cos \theta$$

Thus, the Lagrangian is now given by:

$$\mathcal{L} = \frac{1}{2} (M + m) \dot{x}^2 + \frac{1}{2} m \ell^2 \dot{\theta}^2 - m \ell \dot{x} \dot{\theta} \cos \theta - mgl \cos \theta \quad (14)$$

And the Lagrange's equation of the system can be written as follows:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) - \frac{\partial \mathcal{L}}{\partial q_k} = Q_k \quad (15)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = Q_\theta$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = Q_x$$

We first differentiate with respect to x :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{x}} &= (M + m)\dot{x} - m\ell\dot{\theta}\cos\theta \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) &= (M + m)\ddot{x} - m\ell\ddot{\theta}\cos\theta + m\ell\dot{\theta}^2\sin\theta \\ \frac{\partial \mathcal{L}}{\partial x} &= 0 \end{aligned}$$

The generalized force Q_x associated with the coordinate x is derived using the principle of virtual work:

$$\delta W = \tau_{\text{friction}} \delta x + f \delta x = (-c\dot{x} + f) \delta x = Q_x \delta x \quad (16)$$

$$\Rightarrow Q_x = f$$

because c is now equal to 0.

The equation becomes:

$$(M + m)\ddot{x} - m\ell\ddot{\theta}\cos\theta + m\ell\dot{\theta}^2\sin\theta = f$$

Then we differentiate with respect to θ :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} &= m\ell^2\dot{\theta} - m\ell\dot{x}\cos\theta \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) &= m\ell^2\ddot{\theta} - m\ell\ddot{x}\cos\theta + m\ell\dot{x}\sin\theta \\ \frac{\partial \mathcal{L}}{\partial \theta} &= m\ell\dot{x}\sin\theta + mg\ell\sin\theta \end{aligned}$$

The generalized force Q_θ associated with the coordinate θ is derived using the principle of virtual work:

$$\delta W = \tau_{\text{friction}} \delta \theta = (-b\dot{\theta}) \delta \theta = Q_\theta \delta \theta \quad (17)$$

$$\Rightarrow Q_\theta = 0$$

because b is now equal to 0.

The equation becomes:

$$m\ell^2\ddot{\theta} - m\ell\ddot{x}\cos\theta - mg\ell\sin\theta = 0$$

The resulting equations are:

$$\boxed{\begin{aligned} (M + m)\ddot{x} - m\ell\ddot{\theta}\cos\theta + m\ell\dot{\theta}^2\sin\theta &= f \\ m\ell^2\ddot{\theta} - m\ell\ddot{x}\cos\theta - mg\ell\sin\theta &= 0 \end{aligned}} \quad (18)$$

B - Nonlinear state-space formulation

- State vector:

$$\mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{Bmatrix}$$

- Selecting the control force and disturbance force as inputs:

$$u + d = f$$

- Selecting as output x_1 which represents the cart horizontal displacement and x_3 which corresponds to the pendulum angular position with respect to the vertical axis:

$$\mathbf{y} = \begin{bmatrix} x \\ \theta \end{bmatrix} = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix}$$

Now we explicit $\ddot{x}$ and $\ddot{\theta}$ from the EOMs:

$$\begin{bmatrix} M + m & -m\ell\cos x_3 \\ -m\ell\cos x_3 & m\ell^2 \end{bmatrix} \begin{bmatrix} \dot{x}_2 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} u + d - m\ell x_4^2 \sin x_3 \\ m g \ell \sin x_3 \end{bmatrix} \quad (19)$$

We can as before solve the system by inverting the mass matrix.

Let:

$$D = (M + m)m\ell^2 - (m\ell\cos x_3)^2$$

Then, the accelerations are given by:

$$\begin{cases} \ddot{x} = \dot{x}_2 = \frac{1}{D} [m\ell^2(u + d - m\ell x_4^2 \sin x_3) + m^2 \ell^2 g \cos x_3 \sin x_3] \\ \ddot{\theta} = \dot{x}_4 = \frac{1}{D} [m\ell \cos x_3 (u + d - m\ell x_4^2 \sin x_3) + (M + m)m g \ell \sin x_3] \end{cases} \quad (20)$$

The nonlinear state-space representation of the inverted pendulum system can now be fully expressed in first-order form.

Compact version:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} x_2 \\ \ddot{x}(\mathbf{x}, \mathbf{u}) \\ x_4 \\ \ddot{\theta}(\mathbf{x}, \mathbf{u}) \end{bmatrix} \quad (21)$$

C - Linearized EOM of the system

The aim is to linearize the nonlinear equations of motion, this time around the equilibrium point:

$$\mathbf{x}_{eq} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}_{eq} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

We employ a first-order Taylor series expansion. Given a non-linear function $f(x, u)$, the Taylor expansion of first order around an equilibrium point (x_e, u_e) is:

$$f(x, u) \approx f(x_e, u_e) + \left. \frac{\partial f}{\partial x} \right|_{(x_e, u_e)} (x - x_e) + \left. \frac{\partial f}{\partial u} \right|_{(x_e, u_e)} (u - u_e) \quad (22)$$

The nonlinear dynamics of the inverted pendulum system is given by the state vector in equation 21, considering the acceleration as written in equation 20 .

Near $x_3 = 0$, the trigonometric functions are approximated through the Taylor expansion as:

$$\begin{aligned} \sin x_3 &\approx (x_3 - 0) \\ \cos x_3 &\approx 1 \end{aligned}$$

Furthermore, higher-order terms involving x_4^2 are neglected.

The mass matrix determinant simplifies to the following:

$$D \approx Mm\ell^2$$

Now we can substitute the first-order approximations into the nonlinear equations, and after some passages, we find:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \ddot{x} = \frac{1}{M}(u + d) + \frac{mg}{M}x_3 \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = \ddot{\theta} = \frac{1}{\ell M}(u + d) + \frac{(M+m)g}{\ell M}x_3 \end{cases} \quad (23)$$

D - Linearized state-space formulation

Having just linearized the EOM, we can directly proceed to the derivations of matrices A, B, C and D of state-space formulation.

As mentioned earlier in **Task 1.4**, the second-order system can be expressed in state-space form as:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{y} = \mathbf{g}(\mathbf{x})$$

where the inputs are u and d , and the outputs are x and θ , previously described.

We want to achieve this kind of formulation:

$$\begin{cases} \dot{\mathbf{x}} = A(x - x_e) + B(u - u_e) \\ \mathbf{y} = C(x - x_e) + D(u - u_e) \end{cases} \quad (24)$$

Let us recall that the state vector can then be written as:

$$\dot{\mathbf{x}} = \begin{bmatrix} f_1(\mathbf{x}, u) \\ f_2(\mathbf{x}, u) \\ f_3(\mathbf{x}, u) \\ f_4(\mathbf{x}, u) \end{bmatrix} = \begin{bmatrix} x_2 \\ \ddot{x}(\mathbf{x}, u) \\ x_4 \\ \ddot{\theta}(\mathbf{x}, u) \end{bmatrix}$$

We also recall the expressions of the matrices:

$$\begin{aligned}
A &= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{bmatrix} \bigg|_{x_{eq}} \\
B &= \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \\ \frac{\partial f_3}{\partial u_1} & \frac{\partial f_3}{\partial u_2} \\ \frac{\partial f_4}{\partial u_1} & \frac{\partial f_4}{\partial u_2} \end{bmatrix} \bigg|_{x_{eq}} \\
C &= \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \frac{\partial g_1}{\partial x_3} & \frac{\partial g_1}{\partial x_4} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \frac{\partial g_2}{\partial x_3} & \frac{\partial g_2}{\partial x_4} \end{bmatrix} \bigg|_{x_{eq}} \\
D &= \begin{bmatrix} \frac{\partial g_1}{\partial u_1} & \frac{\partial g_1}{\partial u_2} \\ \frac{\partial g_2}{\partial u_1} & \frac{\partial g_2}{\partial u_2} \end{bmatrix} \bigg|_{x_{eq}}
\end{aligned}$$

In this case, the evaluation at equilibrium is already done during the process of linearization of the EOM, therefore we proceed directly with all the relative derivations. After some straightforward calculations, we derive the final matrices:

$$\begin{aligned}
A &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M+m)g}{M\ell} & 0 \end{bmatrix} \\
B &= \begin{bmatrix} 0 & 0 \\ \frac{1}{M} & \frac{1}{M} \\ 0 & 0 \\ \frac{1}{M\ell} & \frac{1}{M\ell} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}
\end{aligned}$$

Comparison between the state-space formulations

We can observe some changes from the matrices of the system considered in **part 1**.

- **State matrix A:** In the first model, A includes damping terms proportional to the friction coefficients b and c , affecting the second and fourth rows. In the second model, these terms are absent due to the assumption of no friction. Furthermore, the gravitational coupling terms have opposite signs because of the change in the positive direction of the convention for the pendulum angle and the change of the equilibrium point, this time at $\theta = 0$.
- **Input matrix B:** In the first model, B is a 4×1 matrix corresponding to a single input force. In the second model, B becomes a 4×2 matrix to separately account for the control input u and the disturbance d , both of which affect the dynamics of the system in the same way, because they are the two contributing forces of the force f .
- **Output matrix C:** C remains unchanged between the two models. It still selects the cart position x and the pendulum angle θ as system outputs.
- **Feedthrough Matrix D:** D is zero in both models, indicating the absence of direct feedthrough from the input(s) to the output, but it becomes a 2×2 matrix because we now consider two inputs.

2.2. Task 2.2 – Open-loop transfer function

A - Transfer function

We proceed to derive the open loop transfer function from the control input to the pendulum angle:

$$G(s) = \frac{\theta(s)}{u(s)} \quad (25)$$

In order to select the exact input and output, we consider the first column of B and the second row of C in the implementation the following calculation:

$$G(s) = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} (sI - A)^{-1} \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ \frac{1}{M\ell} \end{bmatrix} \quad (26)$$

And we obtain:

$$G(s) = \frac{1}{M\ell s^2 - Mg - mg}$$

To derive the numerical expression we insert the parameters of the inverted pendulum model:

$$G(s) = \frac{4/9}{s^2 - 13.5} = \frac{0.4}{s^2 - 13.5}$$

B - Pole-zero map

To derive the mathematical expression of the zeros and poles, we have to solve for 0 respectively the numerator and the denominator of the transfer function.

- **Zeros:** the numerator is constant, which means that there aren't any zeros.
- **Poles:** to find the poles, we solve this simple equation:

$$s = \pm\sqrt{13.5} \approx \pm 3.6742$$

Using the MATLAB[®] function *pzmap* we generate the following pole-zero map:

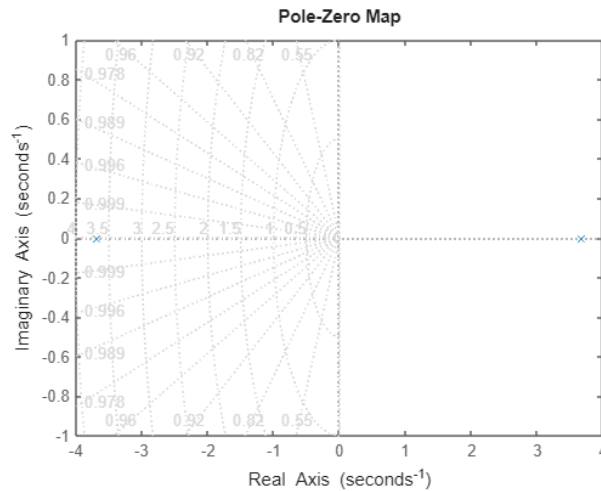


Figure 10: pole-zero map of the linearized system

The transfer function has two real poles, and one of them is positive as we can also observe from the map. This means that the inverted pendulum is an unstable system, as we briefly anticipated at the beginning of **part 2**.

2.3. Task 2.3 – Proportional (P) control for stabilization

A - Closed loop transfer function

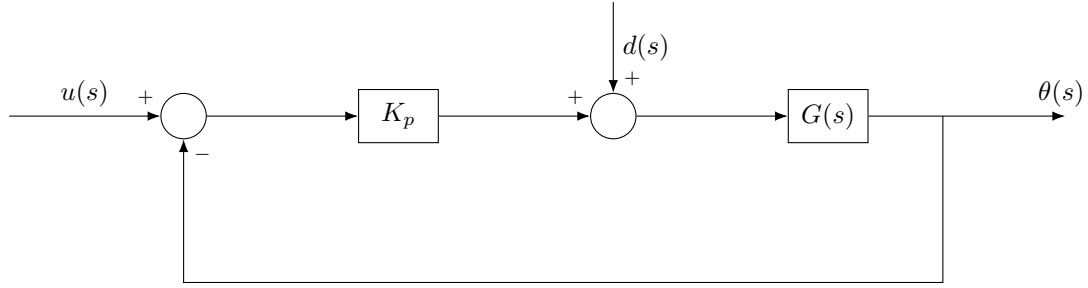


Figure 11: Control scheme with the proportional controller K_p

Having previously derived the instability of the equilibrium with $\theta = 0$, we can try to stabilize the system through a feedback in a closed-loop block diagram. The main objective is to regulate the pendulum angle θ despite external disturbances, using a feedback controller denoted by R which will control the input force f necessary to our end.

The system is based on the feedback control of tracking error:

$$e(s) = r(s) - \theta(s) \quad (27)$$

where $r(s)$ is the reference input (desired angle, in our case 0), and $\theta(t)$ is the actual output (measured angle).

The regulator considered R is a **proportional controller** with gain k_P , and its output is the control force:

$$u(s) = R(s) \cdot e(s) = k_P \cdot e(s) = k_P(r(s) - \theta(s)) = -k_P\theta(s)$$

In particular, the regulator is used to generate an input force proportional to the error which measures the deviation of the actual angle from the desired one, contrasting the tendency of the pendulum to loose its equilibrium.

$$\theta(s) = G(s) [u(s) + d(s)] = G(s) [-k_P\theta(s) + d(s)]$$

where $d(s)$ is an external disturbance acting additively on the input.

After developing and rearranging the expression:

$$\frac{\theta(s)}{d(s)} = G_C^P(s) = \frac{G(s)}{1 + k_P G(s)} \quad (28)$$

B - Verification of the stability

We question ourself if the proportional control can ultimately stabilize the system. We study the position of the poles of the transfer function just obtained:

$$1 + k_P G(s) = 0$$

We replace $G(s)$ with the transfer function we previously calculated:

$$1 + k_P \frac{1}{M\ell s^2 - (M + m)g} = 0$$

This can be rewritten as:

$$s = \pm \sqrt{\frac{(M + m)g - k_P}{M\ell}}$$

We have two different results depending on the value of k_P :

- $k_P < (M + m)g$: the system has two non-null real poles which are symmetric with respect to the Imaginary axis. One of them is negative, and the solution is **unstable**.
- $k_P = (M + m)g$: the system has two coincident poles at the origin, the solution is **marginally stable**.
- $k_P > (M + m)g$: The system has two purely imaginary poles which are symmetric with respect to the Real axis. Both of them have no real part, and the solution is **marginally stable**.

We understand that there is no condition in which the system is stable, but it can reach the status of marginal stability. This means that the system would not have poles with positive real part, which would cause instability, nor it would exhibit exponential decay, this means there would be no poles with negative real part. In such a system, the response might oscillate continuously without growing.

In conclusion, a proportional control cannot stabilize the inverted pendulum system. We must find another solution to stabilize this specific system.

C - Root locus method

To further support what we found in the previously we used the MATLAB[®] function `rlocus` to plot the graph in figure 12.

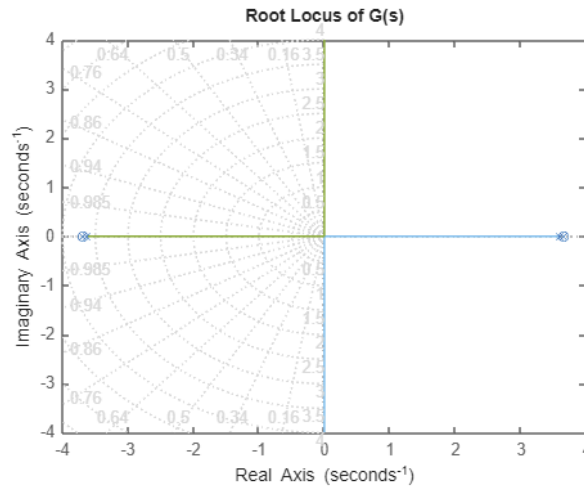


Figure 12: root locus plot of the system

As we can observe the system can only be marginally stable when the poles of the transfer function are purely imaginary or zero. The position of the poles on the graph is dependent to the gain k_p , we will use the previously calculated value of it to define the marginal stability: $k_p = 30.38$.

D - Minimum value of the control gain for marginal stability

As reported in point B, the stability of the system depends on the value of k_p . We saw that for values inferior to $(M + m)g$ the system is unstable, and becomes marginally stable when the control gain equals that point.

Switching to numerical values, the minimum value of the control gain such that the closed-loop system is marginally stable is:

$$k_p = 30.38$$

2.4. Task 2.4 – Proportional-Derivative (PD) control for stabilization

A - Closed loop transfer function

To try to stabilize the system, we now consider a **proportional-derivative** control:

$$R(s) = k_P + sk_D$$

implemented in the following block-scheme:

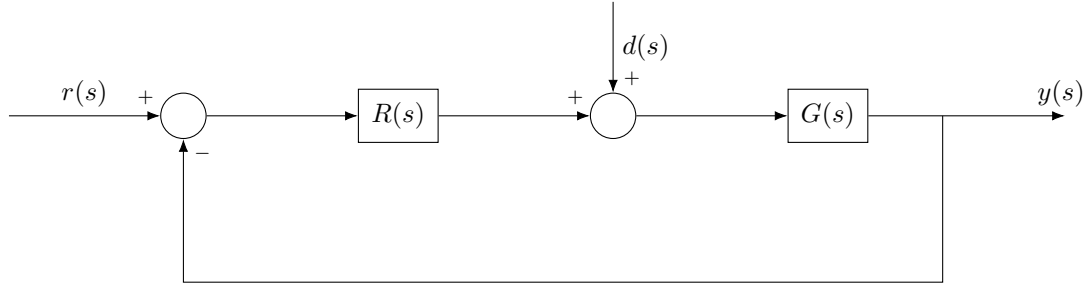


Figure 13: Control scheme of the PD controller $R(s)$

By still assuming our reference equal to 0, we replace in equation 28 k_P with our new expression of the control and we obtain:

$$\frac{\theta(s)}{d(s)} = G_C^{PD}(s) = \frac{G(s)}{1 + (k_P + sk_D)G(s)} \quad (29)$$

If we replace with the expression of $G(s)$ and reformulate we derive the final transfer function from the disturbance to the pendulum angle:

$$G_C^{PD}(s) = \frac{1}{M\ell s^2 + k_D s - (M + m)g + k_P}$$

B - Response to a step disturbance

We now want to apply a step disturbance of $F = 1N$ for $t = 0.1s$ to comply with the specific maximum peak time and overshoot. In control systems, these are key parameters used to describe the time-domain response of a system.

We notice that the denominator of our transfer function matches that of the typical expression of a second-order system, which is given by:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (30)$$

where:

- ω_n is the natural (angular) frequency of the system.
- ζ is the damping ratio.

We can factor out $M\ell$ from the denominator to express it in the standard second-order form:

$$G_C^{PD}(s) = \frac{1}{M\ell} \frac{1}{s^2 + \frac{k_D}{M\ell}s + \frac{k_P - (M+m)g}{M\ell}}$$

By comparing coefficients, we identify the following:

$$2\zeta\omega_n = \frac{k_D}{M\ell}, \quad \omega_n^2 = \frac{k_P - (M+m)g}{M\ell} \quad (31)$$

We can now focus on the parameters object of the study.

The **peak time** represents the time taken for the system response to reach the first maximum after a step input. For a second-order system like ours, it is given by:

$$T_p = \frac{\pi}{\omega_d}, \quad \omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (32)$$

The **overshoot** is the percentage by which the maximal amount of the response exceeds its final steady-state value. It is given by:

$$M_p = 100 \times e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \quad (33)$$

We would like to impose some conditions on these two parameters:

- Peak time under 1 second: $T_p < 1$

$$\omega_n \sqrt{1 - \zeta^2} > \pi \quad (34)$$

- Overshoot under 20%: $M_p < 20$

$$e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} < 0.2 \quad (35)$$

In order to satisfy both these requirements, we must impose constraints on the damping ratio ζ and the natural angular frequency ω_n . This results in solving a system of two coupled nonlinear inequalities.

We first need to satisfy the second inequality (35) since only ζ appears, and we obtain numerically a **minimum condition** as a result.

$$\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} < \ln(0.2) \quad \Rightarrow \quad \zeta > 0.4560$$

In order to be possible to define an overshooting, the system has to be at least underdamped, which means having a damping ratio **less than 1**.

Subsequently, by observing the first inequality (34), we understand that for a constant ω_d , if we increase ζ the peak time decreases rapidly. Therefore the condition on the natural damped angular frequency must be expressed as a function of the choice of the damping ratio.

We rewrite the inequality for the peak time to derive the final pair of inequalities:

$$\begin{cases} 0.4560 < \zeta < 1 \\ \omega_n > \frac{\pi}{\sqrt{1-\zeta^2}} \end{cases}$$

Simultaneously imposing these two conditions ensures to our system a fast and sufficiently damped (with limited oscillatory dynamics) response, aligning its behavior with typical performance requirements encountered in practical control applications.

To determine the value of the control gains k_P and k_D , we replace the recent results into the expressions found (31) to obtain:

$$\begin{cases} k_D > 2\pi M\ell \frac{\zeta}{\sqrt{1-\zeta^2}} \\ k_P > M\ell \frac{\pi^2}{1-\zeta^2} + (M+m)g \end{cases} \quad \text{with } 0.4560 < \zeta < 1 \quad (36)$$

These inequalities define the admissible region in the (k_D, k_P) parameter space that ensures the closed-loop system satisfies the requirements of peak time less than 1 second and overshoot below 20%.

Desired locations of closed-loop poles

These constraints suggest particular desired locations for the poles of our closed-loop transfer function. We recall the denominator to be in the form:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 \rightarrow s^2 + \frac{k_D}{M\ell}s + \frac{k_P - (M+m)g}{M\ell}$$

The poles will then be in the form:

$$p_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

By distinguishing real part and imaginary part, we can apply easily the inequalities:

$$\Re(p) = -\zeta\omega_n < -0.456 \cdot \omega_n, \quad \Im(p) = \omega_n \sqrt{1 - \zeta^2} > \pi$$

By definition the natural angular frequency ω_n must be positive, therefore $\Re(p)$ will always be negative and the poles must be positioned on the left half of the complex plane, so that:

- The real part $-\zeta\omega_n < 0$ ensures **stability** and sufficient damping.
- The imaginary part $\omega_n \sqrt{1 - \zeta^2} > \pi$ ensures a fast response (**peak time under 1 second**).

C - Nonlinear closed-loop time response (Matlab®)

In this task we aim to simulate and analyze the nonlinear closed-loop time response of the inverted pendulum system under the action of the PD controller, when a pulse disturbance force is applied to the cart.

The pendulum starts in the unstable equilibrium position, perfectly upright with zero velocity. The external disturbance is modeled as a 1 N rectangular pulse applied to the cart between $t = 0$,s and $t = 0.1$,s.

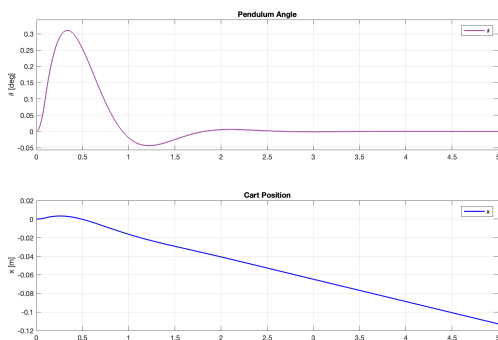
Regarding control gains, we need to select ζ and ω_n to meet the desired transient specifications discussed at **point B**. We choose, for example, $\zeta = 0.5000 > 0.4560$ to obtain (36) the inequalities $k_D > 8.1621$ and $k_P > 59.9888$.

We select, for example, $k_D = 10$ and $k_P = 70$.

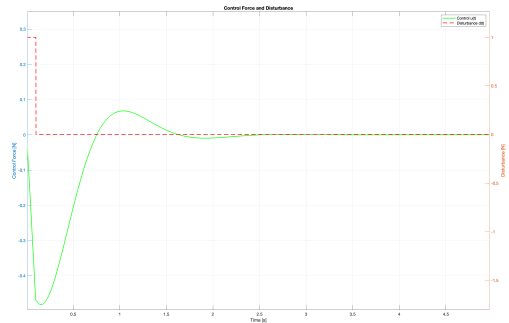
The complete nonlinear dynamic model is implemented in MATLAB® using the ODE solver `ode45`, which integrates the system starting from nonlinear equations of motion (19) The PD controller is applied directly to the pendulum angle θ , with the aim of stabilizing it around $\theta = 0$ even after applying a disturb d thanks to the control force f :

$$u(t) = -k_P \cdot \theta(t) - k_D \cdot \dot{\theta}(t) \quad (37)$$

The simulation was run over a 5-second time horizon, using a fixed time vector with 1 ms spacing to ensure accurate sampling of the short-duration disturbance. Three signals were analyzed: the pendulum angle $\theta(t)$, the cart position $x(t)$, and the control force $u(t)$. The disturbance $d(t)$ was overlaid only on the control subplot for clarity.



(a) Pendulum Angle and Cart Position



(b) Disturb and Control Forces

Figure 14: Nonlinear closed-loop response to a pulse disturbance.

As shown in **Figure 14**, the control system effectively mitigates the effect of the impulsive disturbance and follows the requests for peak-time and overshoot:

- The **pendulum angle** $\theta(t)$ exhibits a small deviation from equilibrium, with a peak of approximately 0.3 degrees. The response is underdamped, but well controlled, returning smoothly toward zero without oscillatory divergence.
- The **cart position** $x(t)$ initially exhibits a slight displacement in the positive direction, directly resulting from the disturbance force. Shortly after, due to the corrective action of the PD controller, the cart quickly inverts its motion and begins to move in the negative direction. In the absence of friction, the cart-pendulum system preserves its linear momentum and moves backward with a constant velocity.
- The **control force** $u(t)$ responds immediately to the disturbance, peaking at approximately -0.5 N shortly after the onset of the pulse, then oscillates and converges to 0 as the angular deviation is corrected. The control action closely mirrors the disturbance profile initially, but persists longer due to the inertia of the system and internal dynamics.

The results confirm that the PD controller, designed using the linearized model, performs effectively when applied to the full non-linear system. The angular deviation remains minimal and stable despite the impulsive disturbance, while the control force remains within moderate bounds. We managed to stabilize the inverted pendulum vertically.

These outcomes validate the robustness of the controller and the suitability of linear design methods for small perturbations around the stable equilibrium.

The limitation of this controller is that it is only a theoretical solution and cannot be implemented to realize a real and functional controller, as the term with k_D indicates a pure derivative control, which requires an instantaneous response to work. There is no actuator or sensor that can offer an instantaneous response in real-life situations thus the non-feasibility of this type of control on the system for a practical use. We will expand this concept further in **Task 2.5A**.

D - Simulink® model

To obtain the non-linear response (cart position and pendulum angle) of the inverted pendulum system under PD control, we developed a Simulink® model based on the previous Task and using an M-function.

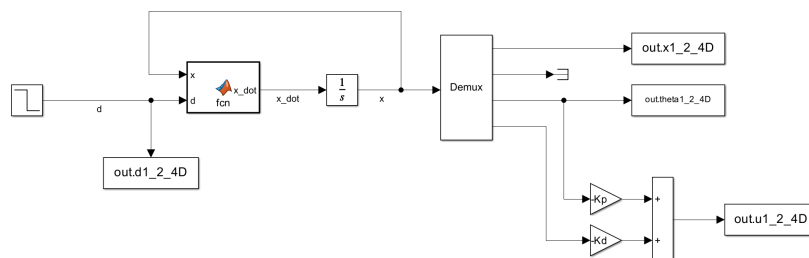
The system employs a function that takes as input the step disturbance force d and computes the first derivative of the non-linear state-space system using the equation (19). This derivative is then integrated, and the resulting state vector x is fed back into the function. All required parameters (including control gains $k_P = 10$ and $k_D = 70$) and matrices are defined within the M-function.

Regarding the settings, the simulation was configured to start at $t = 0.0$ s and end at $t = 5.0$ s. The numerical integration was performed using the **ode45** solver, with a Max step size of 0.01.

As for the model composition, the following blocks were utilized:

- **Step** block (used to apply a 1 N pulse force F from 0 to 0.1 s)
- **Integral** block (configured with the initial conditions $x_0 = [0; 0; 0; 0]$, used to compute the system states over time by integrating their derivatives)
- **Demux** block (used to separate the outputs in order to send x and θ to the workspace)
- **Terminator** blocks (used to terminate $\dot{x}$)
- **Sum** block and two **Gain** blocks (used to obtain $u(t)$)
- **To Workspace** blocks (used to export the outputs to MATLAB®)

The following image displays the Simulink® model described above:



According to the model, the pendulum angle θ , the cart position x and the control force u (with the disturbance force d) were plotted as shown below:

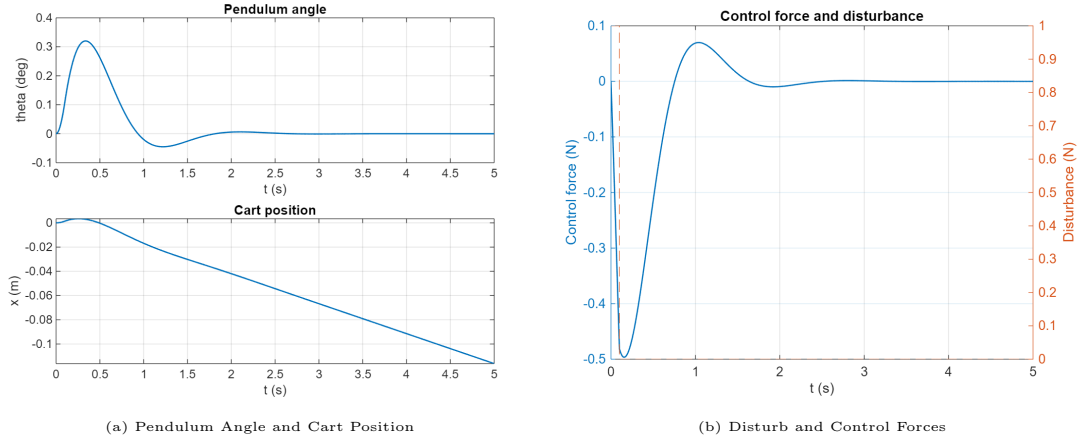


Figure 15: Nonlinear closed-loop response to a pulse disturbance based on Simulink[®] model.

As observed from the plots, the results obtained from the model are consistent with those from the previous task. In particular, the pendulum angle θ shows an underdamped response with a sharp rise after the disturbance followed by decaying oscillations, demonstrating effective stabilization. The cart position x responds with a smooth transient, reaching a maximum displacement of approximately 0.15, m before gradually returning to its initial position without significant oscillations. This indicates a good disturbance rejection. The control force u reacts immediately to the disturbance input, applying a strong corrective action that quickly stabilizes. This confirms the proper adjustment of the PD controller parameters ($k_P = 10$, $k_D = 70$).

Overall, the system becomes stable in about 5 seconds, with minimal overshoot and adequate damping, confirming the effectiveness of the nonlinear control used in the Simulink[®] model.

2.5. Task 2.5 – Partial and full state feedback control

A - Correspondence with a partial-state feedback control

Our aim is to demonstrate that the previous transfer function:

$$G_C^{PD}(s) = \frac{1}{M\ell} \frac{1}{s^2 + \frac{k_D}{M\ell}s + \frac{k_P - (M+m)g}{M\ell}}$$

is in fact a partial-state feedback control.

To this end, we design a partial-state feedback controller that acts only on θ and its velocity, in order to verify that the resulting characteristic equation, and thus the closed-loop poles, remain unchanged.

From the linearized dynamics of the pendulum:

$$\mathbf{x} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}, \quad \dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ \frac{(M+m)g}{M\ell} & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \frac{1}{M\ell} \end{bmatrix} u \quad (38)$$

the previous PD controller was implemented as follows:

$$u = -k_P\theta - k_D\dot{\theta}$$

It controls only the dynamics of the pendulum angle θ (reference for $\mathbf{x}$ equal to $\mathbf{0}$).

$$u = -K\mathbf{x} = -\begin{bmatrix} k_P & k_D \end{bmatrix} \mathbf{x}$$

with system matrices:

$$A = \begin{bmatrix} 0 & 1 \\ \frac{(M+m)g}{M\ell} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{M\ell} \end{bmatrix}$$

The closed-loop system matrix becomes:

$$A_{CL} = A - BK = \begin{bmatrix} 0 & 1 \\ \frac{(M+m)g}{M\ell} - \frac{k_P}{M\ell} & -\frac{k_D}{M\ell} \end{bmatrix}$$

We derive the characteristic equation of the controller:

$$\det(sI - A_{CL}) = \begin{bmatrix} s & -1 \\ \frac{(M+m)g - k_P}{M\ell} & s + \frac{k_D}{M\ell} \end{bmatrix} = s^2 + \frac{k_D}{M\ell}s + \frac{k_P - (M+m)g}{M\ell}$$

This expression matches exactly the denominator of $G_C^{PD}(s)$. Since the controller just designed realizes the same closed-loop dynamics, it confirms that the transfer function in **Task 2.4** was indeed implemented through **partial-state feedback control**.

To find the poles of the transfer function we need to solve the equation above, which gives us the following results:

$$s = -\frac{k_D}{2M\ell} \pm \sqrt{\left(\frac{k_D}{M\ell}\right)^2 - \frac{4(k_P - (M+m)g)}{M\ell}} \quad (39)$$

where the discriminant is defined as:

$$\Delta = \left(\frac{k_D}{M\ell}\right)^2 - \frac{4(k_P - (M+m)g)}{M\ell}$$

This enables us to make the following response classification:

- $\Delta > 0$: 2 real distinct poles $\rightarrow$ overdamped
- $\Delta = 0$: real repeated pole $\rightarrow$ critically damped
- $\Delta < 0$: complex conjugate poles $\rightarrow$ underdamped

It is important to note that the implemented PD controller, although effective in simulation, is not physically realizable in its ideal form. The main limitation stems from the derivative term, which assumes direct and noise-free access to the angular velocity $\dot{\theta}(t)$. In a real-world scenario, the velocity measurement is often derived numerically from position data, introducing high-frequency noise. Then, the derivative action inherently amplifies this noise, which can result in unstable control signals and undesirable actuator behavior.

Moreover, a pure derivative controller implies the use of non-causal operations, requiring future values of the signal, which are obviously not available in real time. Therefore, the implementation of a pure derivative controller is not feasible in hardware without modifications; these modifications would include filtered derivatives

To address these limitations, derivative terms in real systems are typically approximated using filtered derivatives, such as:

$$\dot{\theta}(t) \approx \frac{\tau s}{\tau s + 1} \theta(t)$$

where τ is a small time constant that introduces a low-pass filter to attenuate high-frequency noise.

B - Extension to a full state feedback control

Our challenge now will be to extend the control to a **full-state feedback control**. This means that both the pendulum angle and the cart position will be controlled and stabilized around our reference (which is will be 0 for both), and we will have 4 control gains.

The first step towards the solution is to verify that the control matrix $\mathcal{C}$ of the system has full rank:

$$\text{Rank}(\mathcal{C}) = n$$

where n is the order of the system ($n = 4$).

The control matrix is defined as:

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & A^3B \end{bmatrix} \quad (40)$$

Using the A and B matrices of the linearized system 24 we get:

$$\mathcal{C} = \begin{bmatrix} 0 & \frac{1}{M} & 0 & \frac{mg}{M^2\ell} \\ \frac{1}{M} & 0 & \frac{mg}{M^2\ell} & \frac{m^2g^2}{M^3\ell} \\ 0 & \frac{1}{M\ell} & \frac{(M+m)g}{M^2\ell^2} & 0 \\ \frac{1}{M\ell} & 0 & \frac{(M+m)g}{M^2\ell^2} & \frac{(M+m)^2g^2}{M^3\ell^3} \end{bmatrix}$$

By substituting the parameters of the system we derive the numerical expression:

$$\mathcal{C} = \begin{bmatrix} 0 & 0.3333 & 0 & 0.1452 \\ 0.3333 & 0 & 0.1452 & 0.0474 \\ 0 & 0.4444 & 6.0016 & 0 \\ 0.4444 & 0 & 6.0016 & 81.0804 \end{bmatrix}$$

The matrix $\mathcal{C}$ has $\text{Rank}(\mathcal{C}) = 4$, the system is therefore controllable, and we can proceed towards our objective.

To find the appropriate controller for this system we hypothesize a PD controller on both the pendulum angle θ and the cart position x ; this control would have this structure:

$$u = -\mathbf{K}\mathbf{x} \quad (41)$$

where $\mathbf{K}$ is the array containing the 4 control gains k_i of the system:

$$\mathbf{K} = \begin{bmatrix} k_1 & k_2 & k_3 & k_4 \end{bmatrix}$$

We remind that $\mathbf{x}$ is the vector of the variables:

$$\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$$

For our purposes, we use the Linear-Quadratic Regulator (LQR) design, through the Matlab[®] function `lqr`, which calculates the optimal gain matrix $\mathbf{K}$ taking as inputs the matrices A and B of our system, and the matrices Q and R . In the context of this problem, the matrices Q and R define the trade-off between state regulation and control effort. In particular, Q penalizes deviations of the state variables (the pendulum angle, the cart position and their velocities), while R penalizes the magnitude of the control input (the force f applied on the cart to stabilize the system after the disturbance input).

By appropriately choosing Q and R , we can prioritize either stabilization accuracy or energy efficiency.

For the context of our objectives, we desire to keep the pendulum vertical and the cart still, we assign higher weight to θ and x , as these are the most critical for the stability and performance of the system. The velocities $\dot{x}$ and $\dot{\theta}$ were assigned lower weights, as they are less critical but still relevant for damping and transient behavior.

$$Q = \text{diag}(q_1, q_2, q_3, q_4) = \text{diag}(10, 1, 100, 1)$$

With regard to R , we choose a low value to have both relatively strong control forces when necessary, for faster stabilization, and penalization of excessive control input to ensure efficiency.

$$R = r = 0.01$$

The code returns the values of k_i as:

$$\mathbf{K} = \begin{bmatrix} -31.6228 & -38.7629 & 247.7110 & 63.1943 \end{bmatrix}$$

The plot results are exposed in figure 16:

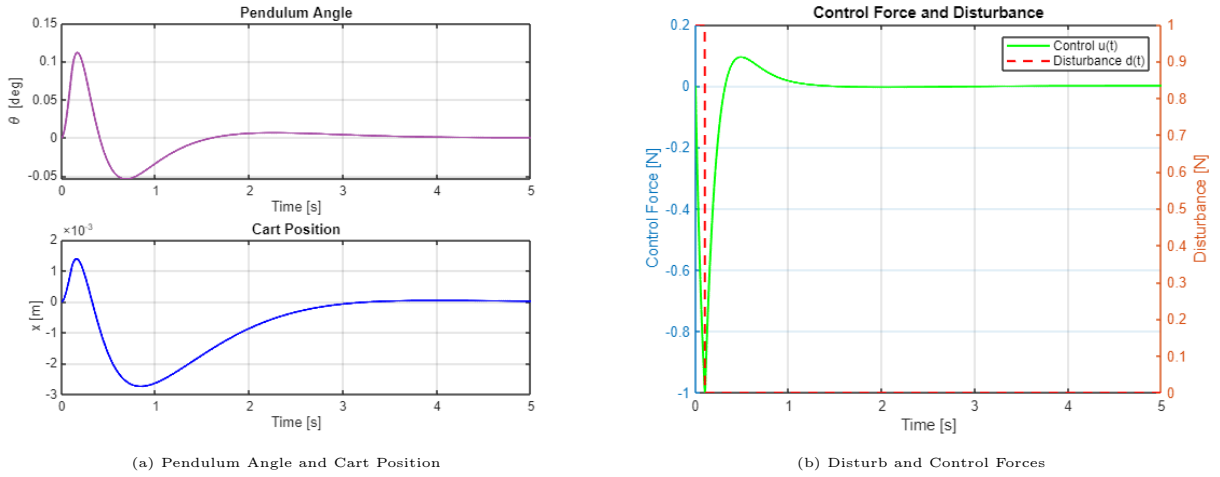


Figure 16: Nonlinear closed-loop response to a pulse disturbance.

As we can see, the effectiveness of the function `lqr` guarantees a very stable and rapid response performance:

- The **pendulum angle** $\theta(t)$ peaks at slightly over 0.1 degrees after almost a tenth of a second, and settles itself around the equilibrium point shortly after.
- The **cart position** $x(t)$ sees an oscillation of the order of the millimeter and returns back to its position to keep it after roughly 3 seconds.
- We can clearly see that the **control force** $u(t)$ peaks this time at -1 N, balancing perfectly the disturbance force. We can presume that this difference from the partial-state feedback control is due to the fact that also the balance of the cart position is needed, so the impulse response of the control force must equal the impulse disturbance in magnitude and be its opposite.

It still persists longer due to the inertia of the system and internal dynamics.

Influence of the selection of the poles on the response of the system

By deriving the eigenvalues of the characteristic polynomial of the closed-loop matrix

$$A_{CL} = A - BK$$

we find the position of the poles for our choice of control gains:

$$\mathbf{P} = \begin{bmatrix} -6.2269 + 3.2762i \\ -6.2269 - 3.2762i \\ -1.3551 + 0.9724i \\ -1.3551 - 0.9724i \end{bmatrix}$$

The real parts of the poles are all negative, so our control PD indeed stabilizes the system.

If we wish to investigate the effects of different selections of closed-loop poles, we can use the Matlab[®] function `place` which for desired poles locations returns the $\mathbf{K}$ vector of the control gains.

For this purpose we evaluate the **control effort** for each choice of poles, which is defined as a measure of the energy of the control force $u(t)$ required. A commonly adopted definition in continuous-time systems is the squared L_2 norm of the control signal:

$$J_u = \int_0^T u(t)^2 dt \quad (42)$$

where T is the time horizon of interest.

For example, our choice of poles implies $J_u = 0.0915 \text{ N}^2 \cdot \text{s}$.

We can try before to select poles closer to the origin, such as

$$\mathbf{P} = \begin{bmatrix} -5 + 2.2i \\ -5 - 2.2i \\ -1 + 0.9i \\ -1 - 0.9i \end{bmatrix}$$

to obtain a control effort of $J_u = 0.0732 \text{ N}^2 \cdot \text{s}$, and then to select poles further to the left (with more negative real parts), such as:

$$\mathbf{P} = \begin{bmatrix} -20 + 10i \\ -20 - 10i \\ -10 + 5i \\ -10 - 5i \end{bmatrix}$$

to obtain this time a control effort of $J_u = 0.1331 \text{ N}^2 \cdot \text{s}$.

It would seem that keeping the poles closer to the origin is a benefit in terms of control effort, but if we try to plot the relative responses, as we see in Fig.17, we understand that this leads also to a slower response of the system.

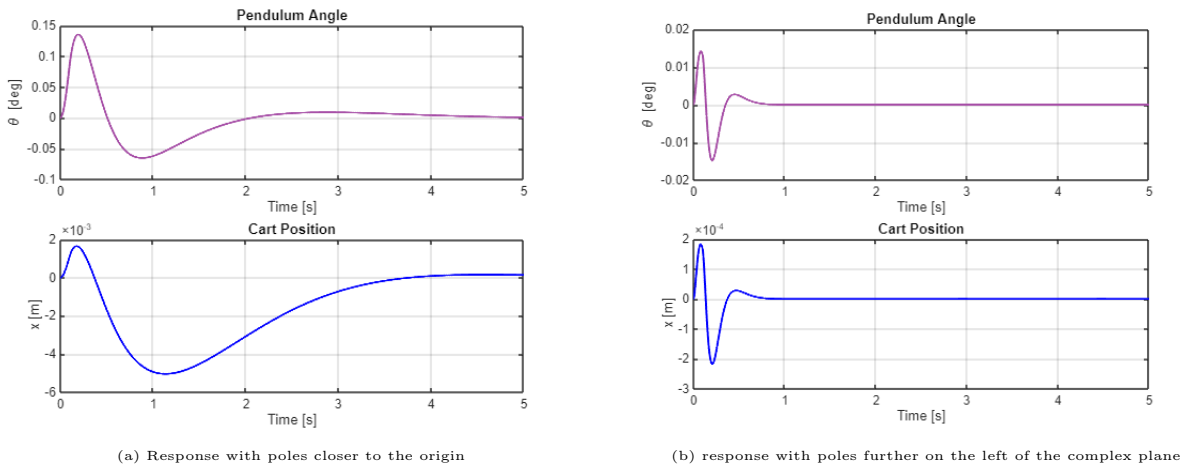


Figure 17: Nonlinear closed-loop response to a pulse disturbance.

We prove even more that the placement of the closed-loop poles plays a critical role in the behavior of a controlled system, particularly in terms of its dynamic response and control effort.

When poles are placed closer to the origin in the complex plane, the system tends to respond more slowly, requiring less control input and thus a lower control effort. On the other hand, placing poles with more negative real parts leads to faster responses but typically comes at the cost of a higher control effort, as the controller must exert stronger and more frequent actions to keep the equilibrium.

Conclusion

We were finally able to stabilize the inverted pendulum system by applying a full-state feedback control. We demonstrated throughout the analysis that not all types of controllers are suitable for an unstable system. Our analysis revealed that there is a matter that one must consider while designing the system, that is how the placement of closed-loop poles affects both system performance and control effort.

An appropriate balance in pole placement must be achieved to ensure desirable dynamic behavior while maintaining an efficient and feasible control input, especially in the presence of physical or energy limitations. Therefore the designer must clearly define both the objectives of the project and the practical limitations of the system.